General targeted machine learning for modern causal mediation analysis

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Introduction

Recently, the definition and identification of causal mediational effects that are guaranteed to measure the *mechanisms* through which the effect operates has been the subject of considerable debate and methodological work, including but not limited to natural direct and indirect effects (NDE, NIE), randomized interventional direct and indirect effects (RIDE, RIIE), and recanting-twins effects.

In spite of important progress in the nonparametric definition and identification of mediational parameters, estimators for many approaches remain underdeveloped, with only a handful that can handle **continuous or high-dimensional mediators** in specific settings. However, methods for mediation analyses that can handle multivariate or high-dimensional mediators are paramount in contemporary research fields such as multi-omics analysis.

Additionally, the literature has almost exclusively focused on binary exposures, although some approaches for continuous and/or multivariate exposures have been developed based on stochastic interventions and modified treatment policies.

Contribution

1. Develop non-parametric double machine learning estimators for six mediation parameters for the case of binary or continuous treatments (A) in the presence of continuous and/or multivariate, mediators (M) and intermediate confounders (Z).
2. Propose two approaches for defining mediation parameters in the presence of continuous or multivariate treatments, and to propose estimation methods that can handle multivariate mediators.

Natural direct and indirect effects (NDE/NIE)
Decision theoretic direct and indirect effects (DTDE/DTIE)
Organic direct and indirect effects (ODE/OIE)
Randomized interventional direct and indirect effects (RIDE/RIIE)
Path-specific effects based on recanting twins (RT)
Path-specific effects based on separable effects (SE)

Table 1: Information of the Six Mediation Parameters

Causal DAG

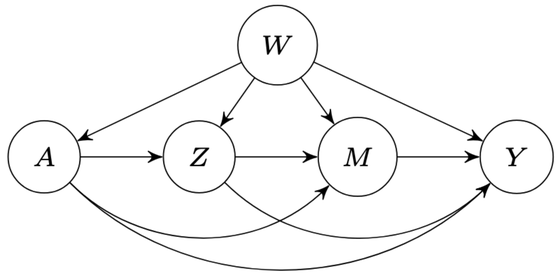


Figure 1: Causal Directed Acyclic Graph (DAG)

Results

		recanting twins					interventional		natural	
n	metrics	AY	AZY	AZMY	AMY	Int. Confounder	IDE	IIE	NDE	NIE
500	Bias	0.016	-0.014	-0.020	0.004	-0.003	-0.005	0.005	0.000	-0.005
	$\sqrt{n}$-Bias	0.386	-0.302	-0.439	0.092	-0.058	-0.102	0.115	0.000	-0.121
	nMSE	9.009	15.79	3.500	4.917	9.948	5.135	11.53	7.189	3.143
	Coverage	0.942	0.940	0.952	0.950	0.960	0.942	0.960	0.954	0.940
1000	Bias	0.023	-0.033	-0.015	0.000	0.003	-0.005	0.002	0.005	-0.004
	$\sqrt{n}$-Bias	0.725	-1.032	-0.464	-0.002	0.103	-0.158	0.052	0.166	-0.128
	nMSE	8.961	10.82	2.333	3.270	5.722	4.416	10.03	5.996	1.306
	Coverage	0.932	0.930	0.942	0.946	0.970	0.950	0.948	0.962	0.942
2000	Bias	0.021	-0.026	-0.014	-0.002	0.005	0.000	0.000	0.003	0.000
	$\sqrt{n}$-Bias	0.951	-1.155	-0.628	-0.077	0.240	0.014	0.016	0.142	0.007
	nMSE	7.709	4.760	1.513	2.276	4.670	3.597	10.123	5.972	1.029
	Coverage	0.948	0.888	0.934	0.958	0.964	0.936	0.940	0.956	0.966

Identification

Define $\theta(A,Z,M,W) = \mathsf{E}[Y \mid A,Z,M,W]$. For values $(a_1,\ldots,a_4) \in \{0,1\}^4$, we define

$$\psi^R(a_1,a_2,a_3,a_4) = \mathsf{E}\Big[\mathsf{E}\Big[\mathsf{E}\Big[\mathsf{E}\big[\theta(a_1,Z^\pi,M,W) \,\big|\, A=a_2,M,W\big] \,\Big|\, A=a_3,Z,W\Big] \,\Big|\, A=a_4,W\Big]\Big],$$
$$\psi^N(a_1,a_2,a_3) = \mathsf{E}\Big[\mathsf{E}\Big[\mathsf{E}\big[\theta(a_1,Z,M,W) \,\big|\, A=a_2,Z,W\big] \,\Big|\, A=a_3,W\Big]\Big], \tag{1}$$

where Z^π satisfying $Z^\pi \mid (A,W) \sim Z \mid (A,W)$ and $Z^\pi \perp\!\!\!\perp M \mid (A,W)$, then the identifications for all six mediation parameters can be written in terms of ψ^N and ψ^R. For instance,

$$\text{NDE} = \psi^N(1,0,0) - \psi^N(0,0,0), \qquad \text{NIE} = \psi^N(1,1,1) - \psi^N(1,0,0),$$
$$\text{RIDE} = \psi^R(1,1,0,0) - \psi^R(0,0,0,0), \qquad \text{RIIE} = \psi^R(1,1,1,1) - \psi^R(1,1,0,0),$$

as well as

$$\text{RT}_1 = \psi^N(1,1,1) - \psi^N(0,1,1), \qquad \text{RT}_2 = \psi^R(0,1,1,1) - \psi^R(0,0,1,1),$$
$$\text{RT}_3 = \psi^R(0,0,1,1) - \psi^R(0,0,1,0), \qquad \text{RT}_4 = \psi^N(0,1,0) - \psi^N(0,0,0),$$

A **sequential expectation** approach can be applied to estimate $\hat{\psi}^N$ and $\hat{\psi}^R$. In sample version, $\hat{Z}^\pi$ can be obtained by **permutation**.

Semi-parametric Efficiency (ψ^R)

Denote φ^R as the first order von Mises expansion of ψ^R, then $\psi(\hat{\mathsf{P}}) - \psi(\mathsf{P}) = -\mathsf{E}_{\mathsf{P}}[\varphi(X;\hat{\mathsf{P}})] + R_2(\hat{\mathsf{P}},\mathsf{P})$, with

$$\begin{aligned}\varphi^R(X;\eta^R) &= \bar{\varphi}^R(X;\eta^R) - \psi^R \\ &= \alpha_4^R(A,Z,M,W)\{Y - \theta_4^R(A,Z,M,W)\} \\ &\quad + \alpha_3^R(A,M,W)\{\theta_4^R(a_1,Z^\pi,M,W) - \theta_3^R(A,M,W)\} \\ &\quad + \alpha_2^R(A,Z,W)\{\theta_3^R(a_2,M,W) - \theta_2^R(A,Z,M)\} \\ &\quad + \alpha_1^R(A,W)\{\theta_2^R(a_3,Z,W) - \theta_1^R(A,W)\} \\ &\quad + \theta_1^R(a_4,W) - \psi^R\end{aligned}$$

where $\alpha_1^R(a,w) = \frac{1(A=a_4)}{\mathsf{P}(a|w)}$;
$\alpha_2^R(a,z,w) = \frac{1(A=a_3)}{\mathsf{P}(a|w)} \frac{\mathsf{P}(z|A=a_4,w)}{\mathsf{P}(z|a,w)}$; $\alpha_3^R(a,m,w) = \frac{1(A=a_2)}{\mathsf{P}(a|w)} \frac{\int_z \mathsf{P}(m|A=a_3,z,w)\mathsf{P}(z|A=a_4,w)}{\mathsf{P}(m|a,w)}$; $\alpha_4^R(a,z,m,w) = \frac{1(A=a_1)}{\mathsf{P}(a|w)} \frac{\mathsf{P}(z|A=a_2,w)}{\mathsf{P}(z|a,w)} \frac{\int_z \mathsf{P}(m|A=a_3,z,w)\mathsf{P}(z|A=a_4,w)}{\mathsf{P}(m|a,z,w)}$, and
$\theta_4^R(a,z,m,w) = \mathsf{E}(Y \mid A=a,Z=z,M=m,W=w)$
$\theta_3^R(a,m,w) = \mathsf{E}[\theta_4^R(a_1,Z^\pi,M,W) \mid A=a,M=m,W=w]$
$\theta_2^R(a,z,w) = \mathsf{E}[\theta_3^R(a_2,M,W) \mid A=a,Z=z,W=w]$
$\theta_1^R(a,w) = \mathsf{E}[\theta_2^R(a_3,Z,W) \mid A=a,W=w]$

Estimation (ψ^R)

To estimate $\varphi^R(X;\eta^R)$, we need to estimate $\theta^R = (\theta_i^R)_{i=1}^4$ and $\alpha^R = (\alpha_i^R)_{i=1}^4$. θ^R can be obtained by taking **sequential expectations**, α^R can be obtained by exploiting the **Riesz representation theorem**. This means the loss functions for $(\alpha_1^R,\alpha_2^R,\alpha_3^R,\alpha_4^R)$ may be expressed as

$$\alpha_1^R = \arg\min_\alpha \mathsf{E}[\alpha(A,W)^2 - 2\theta_1^R(a_4,W)],$$
$$\alpha_2^R = \arg\min_\alpha \mathsf{E}\{\alpha(A,Z,W)^2 - 2\alpha_1^R(A,W)\theta_2^R(a_3,Z,W)\},$$
$$\alpha_3^R = \arg\min_\alpha \mathsf{E}\{\alpha(A,M,W)^2 - 2\alpha_2^R(A,Z,W)\theta_3^R(a_2,M,W)\},$$
$$\alpha_4^R = \arg\min_\alpha \mathsf{E}\{\alpha(A,Z,M,W)^2 - 2\alpha_3^R(A,M,W)\theta_4^R(a_1,Z^\pi,M,W)\},$$

and the optimization problems are solved sequentially, using deep learning.

Note that the estimation of **density ratios** in α^R are NOT needed to solve the optimization problems.

Other Properties

The estimator satisfies **asymptotic normality** and **double robustness** under mild assumptions. Refer to the paper for more info.

FYI

Paper link: https://arxiv.org/pdf/2408.14620. An R package `crumble` is available on CRAN at https://github.com/nt-williams/crumble.